

■■ Food Waste Prediction System

Viva Questions & Simple Answers

Project: Machine Learning Based Food Waste Prediction System

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Project at a Glance

Item	Details
Inputs	Food Prepared (kg), People Served, Temperature (°C)
Output	Predicted Food Waste (kg)
Application	Streamlit web application
Visualization	Plotly gauge chart
MAE	0.3568 kg
R ² Score	0.9157
Model file	food_waste_model.pkl

1. What is the title of your project?

The title of our project is **Food Waste Prediction System**.

2. What is the main purpose of this project?

The main purpose is to predict the amount of food waste using Machine Learning and help with better food preparation and planning.

3. Why did you choose this topic?

Food waste is an important practical problem. Predicting expected waste can support better planning and reduce unnecessary preparation.

4. What is Machine Learning?

Machine Learning is a branch of Artificial Intelligence in which a computer learns patterns from data and uses those patterns to make predictions or decisions.

5. What type of Machine Learning problem is this?

It is a **supervised learning regression problem** because the model predicts a numerical value: food waste in kilograms.

6. What are the inputs to your model?

The inputs are **Food Prepared (kg)**, **People Served**, and **Temperature (°C)**.

7. What is the output of the model?

The output is the predicted amount of food waste in kilograms.

8. What is the target variable?

The target variable is the amount of food wasted, measured in kilograms.

9. What is the role of the dataset?

The dataset contains historical examples used to learn the relationship between the input factors and food waste.

10. What is data preprocessing?

Data preprocessing means preparing the data before model training, such as checking values, selecting relevant features, and converting the data into a suitable form for the model.

11. What is a trained model?

A trained model is a Machine Learning model that has learned patterns from the training data and can then be used to make predictions on new inputs.

12. What is food_waste_model.pkl?

It is the saved trained Machine Learning model. The Streamlit application loads this file and uses it to generate predictions.

13. Why did you use Python?

Python has a large ecosystem for data analysis, Machine Learning, visualization, and web application development, making it suitable for this project.

14. Why did you use Pandas?

Pandas is used for loading, organizing, analyzing, and processing tabular data such as the food waste dataset.

15. Why did you use NumPy?

NumPy provides numerical operations and efficient handling of numerical data used in data science workflows.

16. Why did you use Scikit-learn?

Scikit-learn provides Machine Learning algorithms and evaluation tools needed to train and assess the model.

17. Why did you use Streamlit?

Streamlit allows a Python Machine Learning model to be converted into an interactive web application quickly and easily.

18. Why did you use Plotly?

Plotly is used to create the interactive gauge visualization that shows the predicted food waste.

19. What is MAE?

MAE stands for **Mean Absolute Error**. It measures the average absolute difference between actual values and predicted values. Lower MAE is generally better.

20. What is the MAE of your model?

The reported MAE is **0.3568 kg**. This means the average absolute prediction error on the evaluated data is about 0.3568 kg.

21. What is R² Score?

R², or the coefficient of determination, indicates how well the model explains variation in the target variable. A value closer to 1 generally indicates stronger explanatory performance.

22. What is the R² Score of your model?

The reported R² Score is **0.9157**.

23. Is an R² Score of 0.9157 good?

It indicates strong performance on the evaluated dataset. However, model quality should also be judged using appropriate validation and real-world testing, not only one metric.

24. Why should food waste not be negative?

Food waste represents a physical quantity, so a negative value is not meaningful. The application therefore prevents negative displayed predictions by limiting the result to zero or above.

25. What happens when the user clicks Predict?

The application collects the three input values, sends them to the trained model, receives the predicted waste value, and displays the result with a classification and visualization.

26. What are Low, Moderate, and High waste levels?

They are simple categories used by the application to make the numerical prediction easier to interpret.

27. What is the project architecture?

The flow is: **User Input** → **Input Processing** → **Trained ML Model** → **Prediction** → **Classification** → **Plotly Visualization**.

28. How is the application deployed?

The source code is hosted on GitHub and the Streamlit application is deployed through Streamlit Community Cloud.

29. What is GitHub's role in the project?

GitHub stores and manages the project source code and makes it easy to connect the code repository with the deployment platform.

30. What is requirements.txt?

It lists the Python libraries required by the application, such as Pandas, NumPy, Scikit-learn, Streamlit, and Plotly.

31. What are the main features of your application?

It provides food waste prediction, model evaluation metrics, input validation, waste-level classification, and an interactive Plotly gauge chart.

32. What are the benefits of this system?

It can support better food planning, provide quick estimates, reduce unnecessary preparation, and demonstrate a practical Machine Learning application.

33. What are the limitations of your project?

Prediction quality depends on the dataset. The current system uses a limited number of input factors, while real-world food waste can depend on many additional conditions.

34. How can you improve the project in the future?

We can add more real-world data and features, compare multiple algorithms, improve validation, store prediction history, add more visualizations, and provide downloadable reports.

35. What did you learn from this project?

I learned how to work with a dataset, use a trained Machine Learning model, evaluate predictions, build an interactive Streamlit application, visualize results, and deploy the project online.

36. What makes your project different from a simple calculator?

A calculator uses fixed formulas, whereas this system uses a trained Machine Learning model that learns patterns from historical data to estimate food waste for new inputs.

37. Can this project be used in real restaurants or canteens?

It could be adapted for such settings after collecting representative real-world data and validating the model carefully for the specific environment.

38. What would you do if the model accuracy is poor?

I would inspect the data quality, add relevant features, try suitable algorithms, tune model parameters, improve preprocessing, and use proper validation.

39. Why is model evaluation important?

Evaluation helps determine how well the model performs and whether its predictions are useful on data that was not simply memorized during training.

40. Give a short explanation of your entire project.

Our project predicts food waste using Machine Learning. The user enters food prepared, people served, and temperature. The trained model predicts food waste in kilograms. The Streamlit application displays the prediction, waste category, evaluation metrics, and an interactive Plotly chart. The application is deployed online for easy access.

Quick Presentation Tip

For viva answers, keep your response short and confident. Start with the direct answer, then add one sentence of explanation. For numerical metrics, remember: **MAE = 0.3568 kg** and **R² = 0.9157**.